A SELF-STUDY & TEACHING DECK

Quantum Imaginary-Time Evolution

QITE

$$\frac{e^{-\tau H}|\psi_0\rangle}{\|e^{-\tau H}|\psi_0\rangle\|} \longrightarrow |E_0\rangle$$

Ground-state filtering → measured linear algebra → quantum circuits

Worked examples • Molecular hydrogen • QITE–Krylov benchmark design

Five questions organize the whole topic

01 Why does it work?

Relative suppression of excited-state amplitudes.

02 What can a circuit do?

A state-dependent unitary, not the universal filter.

03 What must be learned?

A real linear system from Pauli expectations.

04 Where can it fail?

Missing directions, noise, conditioning, and depth.

5 How do we compare it fairly? Match the scientific problem and count the full cost.

The minimum quantum vocabulary

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle, \qquad |\alpha|^2 + |\beta|^2 = 1$$

$$H|E_n\rangle = E_n|E_n\rangle, \qquad E_0 \leq E_1 \leq \cdots$$

A STATE

A normalized vector. N qubits have 2^N basis amplitudes.

A HAMILTONIAN

A Hermitian energy operator. Its eigenvectors define energy levels.

AN EXPECTATION

An average estimated over repeated state preparations.

The molecular objective is easy to state

Find the ground-state energy of the chosen molecular model.

$$H = \sum_{\alpha=1}^{L} h_\alpha P_\alpha, \qquad P_\alpha \in \{I, X, Y, Z\}^{\otimes N}$$

MODEL

Geometry, basis, active space, electron and spin sector.

START

An explicitly defined initial state, often Hartree–Fock.

REFERENCE

FCI or exact diagonalization in the same finite model.

[6] Google Quantum AI: OpenFermion electronic-structure tutorial (accessed 8 Oct 2026).

Real time changes phases; imaginary time filters

REAL TIME

Populations stay fixed. Only phases rotate.

$$e^{-itH}: \quad c_n \longmapsto c_n e^{-itE_n}$$

IMAGINARY TIME

Relative populations change. Lower energies gain weight.

$$e^{-\tau H}: \quad c_n \longmapsto c_n e^{-\tau E_n}$$

The useful quantity is an energy difference—not the arbitrary zero of energy.

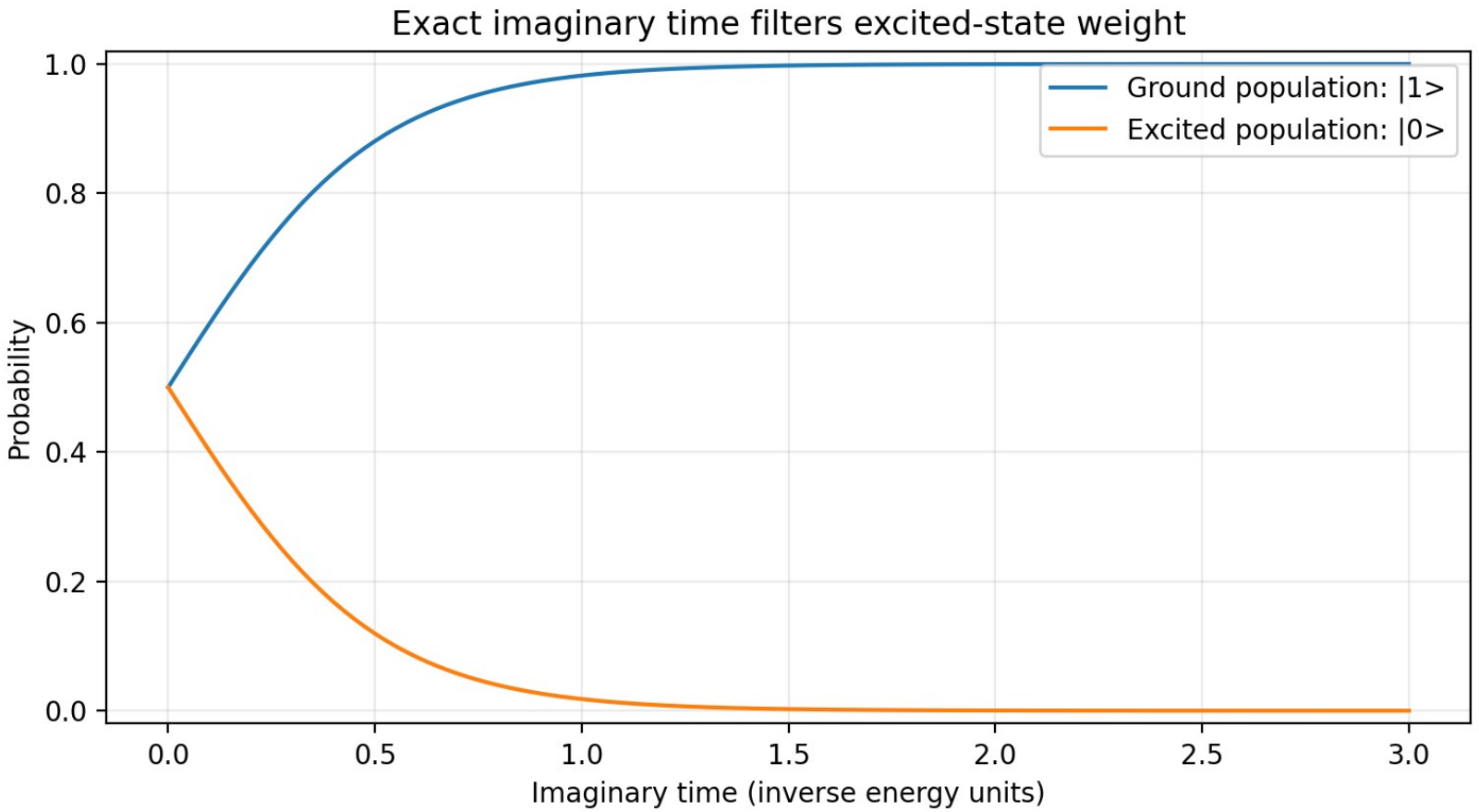
Normalize the filter before interpreting the state

$$|\psi(\tau)\rangle = \frac{e^{-\tau H}|\psi_0\rangle}{\sqrt{\langle\psi_0|e^{-2\tau H}|\psi_0\rangle}}$$

$$p_n(\tau) = \frac{|c_n|^2 e^{-2\tau E_n}}{\sum_m |c_m|^2 e^{-2\tau E_m}}$$

Amplitude ratios decay with exp(−τΔE); population ratios with exp(−2τΔE).

Watch the excited-state weight disappear



H = Z

Start: |+⟩

Ground: |1⟩

Energy: −1

Original calculation for this guide. Exact statevectors; no device noise, shots, or gate-synthesis error.

Convergence has conditions—not a universal promise

$$1 - F_0(\tau) \leq \frac{1 - p_0}{p_0} e^{-2\tau\Delta}, \qquad \Delta = E_1 - E_0 > 0$$

OVERLAP

A ground-space component must be present initially.

GAP

A small relevant gap requires longer imaginary time.

SECTOR

Use the correct electron number and symmetry sector.

Degenerate ground energy → projection into the ground space, not a unique eigenvector.

Why not implement exp(−τH) as an ordinary circuit?

$$(e^{-\tau H})^{\dagger} e^{-\tau H} = e^{-2\tau H} \neq I \quad \text{(generically)}$$

ORDINARY CIRCUIT

One unitary must preserve inner products for all input states.

NORMALIZED FILTER

The normalization depends on the input state; the map is nonlinear.

Non-unitarity is not “forbidden physics”: measurement-based protocols have different costs.

The QITE unitary is tailored to the current state

$$U_{\psi,H}|\psi\rangle \approx \frac{e^{-\Delta\tau H}|\psi\rangle}{\|e^{-\Delta\tau H}|\psi\rangle\|}$$

WHAT IS MATCHED?

The action on one current state—not on every possible input.

WHAT MUST BE RELEARNED?

The generator after the state changes, usually at each update.

State dependence is the way around the universal-unitary obstruction.

[1] Motta et al., Nature Physics 16, 205–210 (2020); arXiv:1901.07653.

Normalization gives a centered flow

$$\partial_\tau |\psi\rangle = -(H - E)|\psi\rangle, \qquad E = \langle\psi|H|\psi\rangle$$

H |ψ⟩

Contains both state-changing and norm-changing components.

(H − E) |ψ⟩

Removes the component that only changes normalization.

A scalar energy offset changes the energy label, not the normalized trajectory.

Exact imaginary time cannot increase energy

$$\frac{\mathrm{d}E}{\mathrm{d}\tau} = -2\left(\langle H^2\rangle - \langle H\rangle^2\right) = -2\,\mathrm{Var}(H) \le 0$$

LARGE VARIANCE

The ideal state is still changing strongly.

ZERO VARIANCE

An energy eigenstate—possibly an excited state.

NOISY ENERGY

Small upward fluctuations do not alone prove a bug.

An exact identity is a diagnostic baseline, not a promise about each noisy data point.

Match the first-order state change

NORMALIZED TARGET

$$|\psi\rangle - \Delta\tau(H - E)|\psi\rangle + O(\Delta\tau^2)$$

UNITARY UPDATE

$$e^{-i\Delta\tau A}|\psi\rangle = |\psi\rangle - i\Delta\tau A|\psi\rangle + O(\Delta\tau^2)$$

$$iA|\psi\rangle \approx (H - E)|\psi\rangle$$

The generator pool is not the Hamiltonian term list

$$A = \sum_{\mu=1}^{M} a_\mu \sigma_\mu, \qquad a_\mu \in \mathbb{R}$$

HAMILTONIAN STRINGS

What determines the energy?

GENERATOR STRINGS

What directions can the state move in?

Example: H = Z and |ψ⟩ = |+⟩ require A = Y, not A = Z.

[1] Motta et al., Nature Physics 16, 205–210 (2020); arXiv:1901.07653.

Measurements determine a real linear system

$$\min_{\mathbf{a}\in\mathbb{R}^M} \left\| (H-E)|\psi\rangle - i\sum_\mu a_\mu \sigma_\mu |\psi\rangle \right\|^2$$

$$S\mathbf{a} = \mathbf{b}$$

$$S_{\mu\nu} = \mathrm{Re}\langle \sigma_\mu \sigma_\nu \rangle,$$

$$b_\mu = \mathrm{Im}\langle \sigma_\mu H \rangle.$$

Convention used here: U = exp(−i Δτ A). The sign of b is tied to this choice.

[1] Motta et al., Nature Physics 16, 205–210 (2020); arXiv:1901.07653.

The Gram matrix can be singular for a good reason

$$\mathbf{x}^T S \mathbf{x} = \left\| \sum_\mu x_\mu \sigma_\mu |\psi\rangle \right\|^2 \geq 0$$

REDUNDANCY

Different operator coefficients can produce the same state motion.

CONDITIONING

Tiny eigenvalues can turn modest data noise into large updates.

REPAIR

Inspect the spectrum. Truncate or regularize; do not invert blindly.

One qubit: calculate the first QITE step by hand

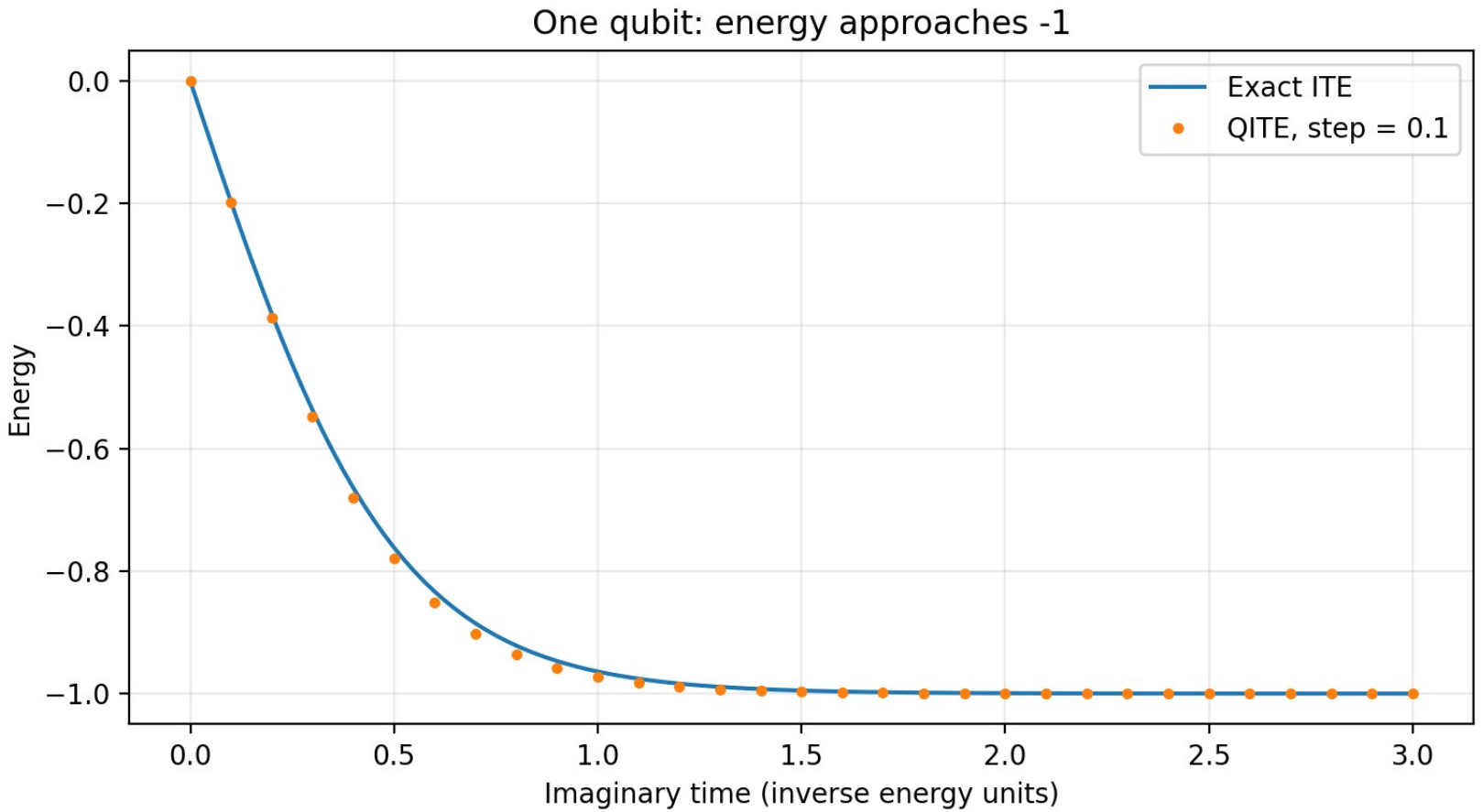
$$H = Z, \qquad |\psi_0\rangle = |+\rangle, \qquad S = I_3, \quad \mathbf{b} = (0, 1, 0)^T$$

$$A = Y, \qquad U = e^{-i\Delta\tau Y} = R_y(2\Delta\tau)$$

AT Δτ = 0.1

QITE: −0.198669 | Exact ITE: −0.197375

The one-qubit energy tends to −1



$$E_{\text{exact}}(\tau) = -\tanh(2\tau)$$

Before any molecule:

Check the sign.

Check normalization.

Recompute A.

Original calculation for this guide. Exact statevectors; no device noise, shots, or gate-synthesis error.

A local pool can miss an entangling direction

$$H = -Z_0 - Z_1 + 0.6X_0X_1, \qquad |\psi_0\rangle = |00\rangle$$

$$(H - E)|00\rangle = 0.6|11\rangle$$

ONE-QUBIT-ONLY POOL

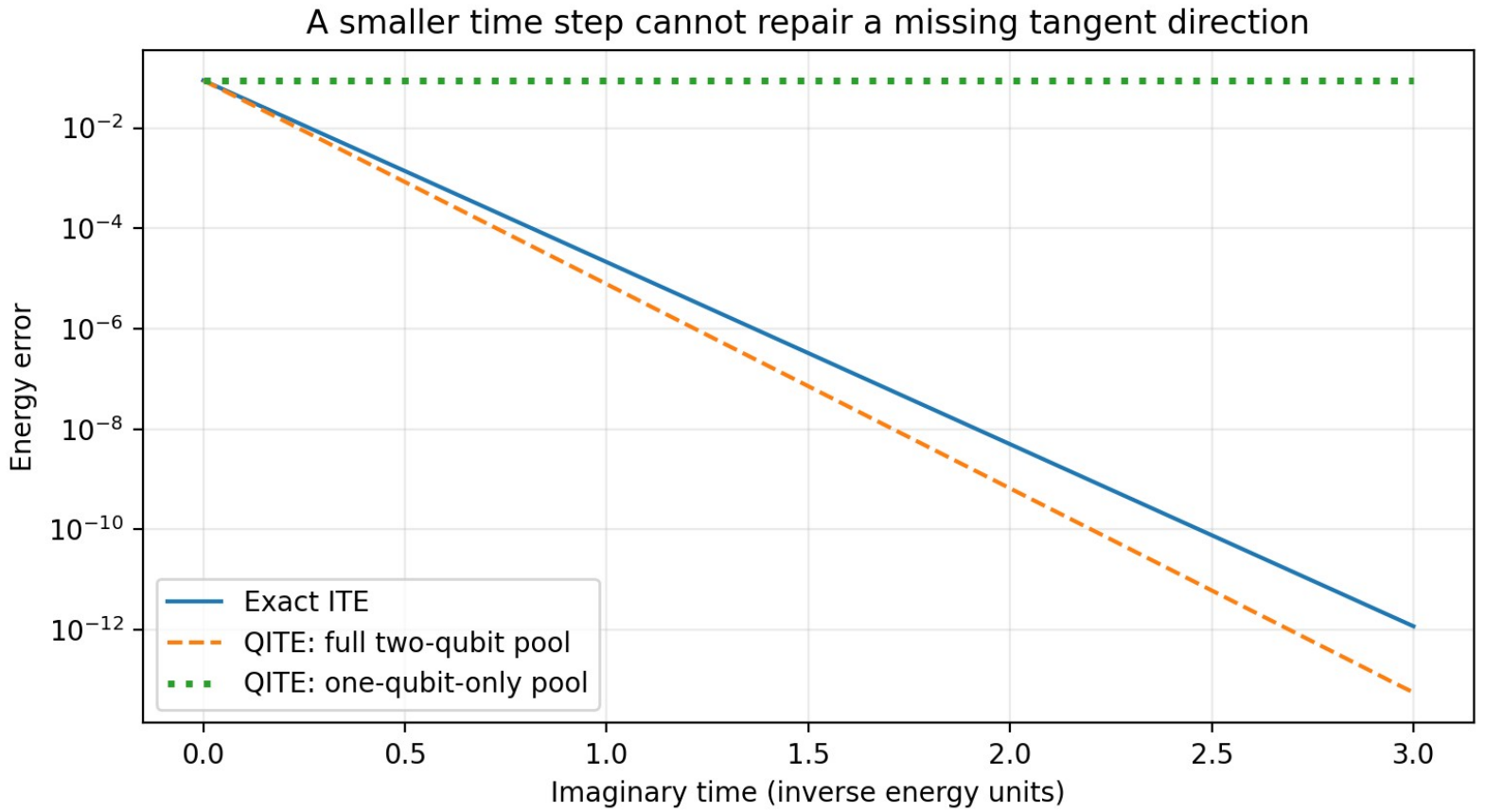
No $|11\rangle$ tangent direction. The update stalls.

A TWO-QUBIT GENERATOR

Provides the missing direction.

$$A = -0.3(X_0Y_1 + Y_0X_1)$$

Smaller steps cannot repair missing directions



Original calculation for this guide. Exact statevectors; no device noise, shots, or gate-synthesis error.

Three approximations must be tracked separately

$$e^{-\Delta\tau \sum_\ell h_\ell} \approx e^{-\Delta\tau h_L} \cdots e^{-\Delta\tau h_1}$$

1 SPLIT H

Approximate one exponential by a product of term filters.

2 FIT THE STATE

Replace each normalized filter by a fitted unitary.

3 SYNTHESIZE U

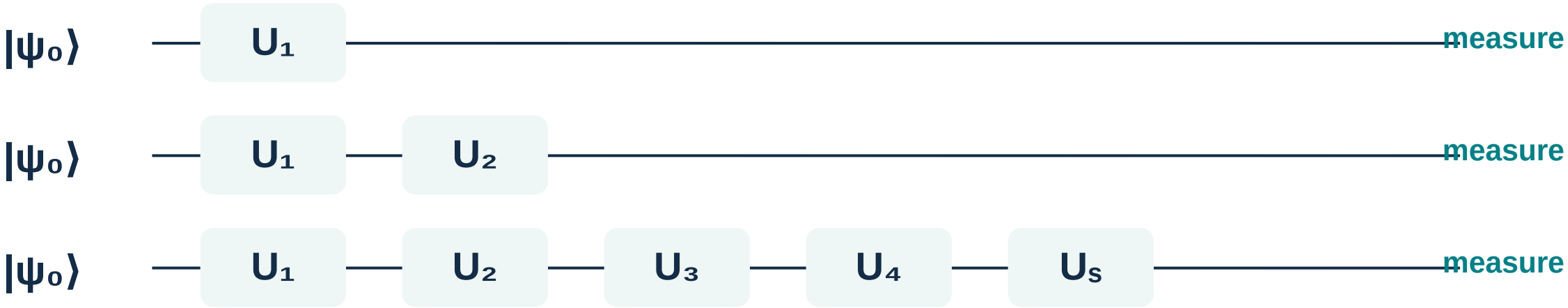
Compile the generator exponential into native gates.

A sweep through L terms advances τ by Δτ—not by LΔτ.

[1] Motta et al., Nature Physics 16, 205–210 (2020); arXiv:1901.07653.

The full preparation circuit usually grows

$$|\psi_s\rangle = U_s \cdots U_2 U_1 |\psi_0\rangle$$



Each shot generally pays for the entire current-state preparation.

Pauli products turn the fit into measurable averages

$$YZ = iX \quad \Longrightarrow \quad \mathrm{Im}\langle YZ \rangle = \langle X \rangle$$

$$\mathrm{SE}(\widehat{p}) = \sqrt{\frac{1 - p^2}{n}} \leq \frac{1}{\sqrt{n}}$$

MEASUREMENT PLAN

Reduce products, remove duplicates, group compatible strings.

COST PLAN

Matrix entries ≠ distinct settings ≠ total circuit executions.

Stabilize the solve before trusting the trajectory

INSPECT

Symmetrize S. Check its spectrum and retained rank.

REGULARIZE

Truncate unresolved directions or penalize large coefficients.

VALIDATE

Test step size and pool separately; evaluate the final state independently.

$$(S + \lambda I)\mathbf{a} = \mathbf{b} \qquad r = \|(H - E)\psi - iA\psi\|$$

A flat energy can mean success—or an inadequate pool, wrong sector, or excessive regularization.

The hybrid loop in one slide

1 PREPARE

Repeated copies of the current state.

→

2 MEASURE

Pauli expectations for S and b.

→

3 SOLVE

Regularized linear system.

→

4 UPDATE

Compile and append $\exp(-i\Delta\tau A)$.

After each sweep: measure total energy, log costs, and test the stopping rule.

[1] Motta et al., Nature Physics 16, 205–210 (2020); arXiv:1901.07653.

Chemistry conventions must match before algorithms do

Geometry + basis + sector

Integrals + HF state

Mapping + optional tapering

Identical qubit Hamiltonian

$$H_{\text{total}} = H_{\text{elec}} + E_{\text{nuc}} I = \sum_{\alpha} h_{\alpha} P_{\alpha}$$

[6] Google Quantum AI: OpenFermion electronic-structure tutorial (accessed 8 Oct 2026).

Hydrogen: an explicit test case, not a performance claim

MODEL

H_2, 0.8 Å
STO-3G
4 qubits, Jordan–Wigner

INITIAL STATE

Hartree–Fock: $|1100\rangle$
Leftmost label = q_0
Nuclear offset included

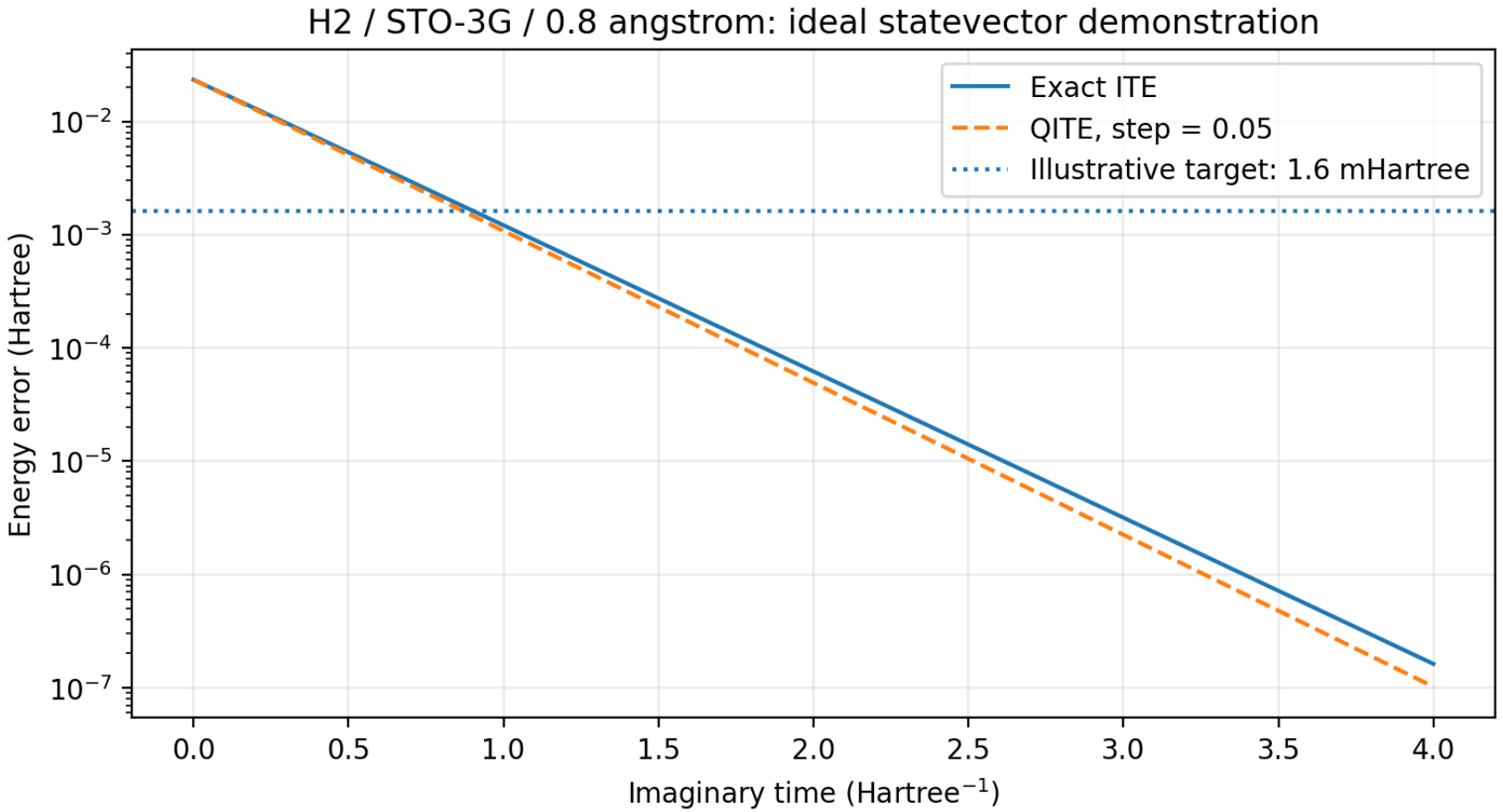
SIMULATION

Full-Hamiltonian QITE
120 odd-Y strings
$\Delta\tau = 0.05$ Hartree^{-1}

$$E_{\mathrm{HF}} = -1.110850397477\, E_{\mathrm{h}} \qquad E_{\mathrm{FCI}} = -1.134147666677\, E_{\mathrm{h}}$$

[6] OpenFermion | Original ideal-statevector calculation; not hardware data

Hydrogen converges in the ideal teaching simulation



80 updates

$\tau = 4$ Hartree^{-1}

Final error: 1.02×10^{-7} Hartree

No hardware conclusion follows.

Original calculation for this guide. Exact statevectors; no device noise, shots, or gate-synthesis error.

The code mirrors the mathematics

```
B = (operators @ psi).T
S = (B.conj().T @ B).real
b = (B.conj().T @ (H @ psi)).imag

a = spectral_pseudoinverse_solve(S,b)
A = sum (a[m]*operators[m] for m in range(M))
psi= expm (-1j*dt*A)@ psi
```

First pass the one-qubit sign test. Then reproduce the toy failure and the H_2 reference.

“QITE” can refer to different algorithm families

LOCAL-UNITARY QITE

Fit A for the current state. Append a unitary. Domain and pool matter.

VARIATIONAL QITE

Fit parameter velocities inside a chosen ansatz. Expressivity matters.

VQE

Optimize circuit parameters for low energy. No trajectory is required.

$$\sum_j M_{ij}\dot{\theta}_j = C_i$$

[2] McArdle et al. | [3] Yuan et al. | [7] Qiskit Algorithms

Quantum Krylov builds a subspace, then diagonalizes

$$|\phi_j\rangle = e^{-it_j H}|\psi_0\rangle$$

$$O_{ij} = \langle\phi_i|\phi_j\rangle, \qquad \mathcal{H}_{ij} = \langle\phi_i|H|\phi_j\rangle$$

$$\mathcal{H}\mathbf{c} = \varepsilon O\mathbf{c}$$

[4] Cortes & Gray, Quantum Krylov subspace algorithms; arXiv:2109.06868.

Krylov has its own conditioning problem

ESTIMATE

Hermitian projected Hamiltonian and overlap matrices.

TRUNCATE

Remove unresolved overlap eigenmodes; record effective rank.

WHITEN

Transform the retained problem into an ordinary eigensolve.

$$T = W_{\mathrm{keep}}\,\mathrm{diag}(d_j^{-1/2}), \qquad H_{\mathrm{white}} = T^{\dagger}\mathcal{H}T$$

An estimated Ritz energy is not automatically a hardware-prepared Ritz state.

[5] Epperly, Lin & Nakatsukasa, SIAM J. Matrix Anal. Appl. 43, 1263–1290 (2022).

Distinct approaches—but not opposites

QITE

State trajectory
Repeated tangent solves
Pool / domain bias
Accumulated update circuit

REAL-TIME KRYLOV

Prepared basis states
Projected eigensolve
Overlap conditioning
Matrix reconstruction cost

Connection: selected QITE states can seed a quantum Lanczos subspace.

[1] Motta et al. | [4] Cortes & Gray

A fair benchmark separates four sources of difficulty

01 Ideal

Validate against the same exact finite-model reference.

02 Finite shots

Isolate sampling error in the fit and matrix reconstruction.

03 Noisy simulation

Use the same justified, documented device assumptions.

04 Hardware

Record device, versions, calibration window, and native costs.

Freeze the Hamiltonian, sector, starting-state policy, and output task first.

Equal shots alone do not imply equal resources

$$N_{\text{shots}} = \sum_c n_c, \qquad G_2^{\text{total}} = \sum_c n_c g_{2,c}, \qquad D_{\max} = \max_c d_c$$

INCLUDE

Training, evaluation, mitigation, calibration, and failed attempts.

SEPARATE

Peak depth, total gates, device time, classical time, and waiting.

VERIFY

Use the chosen backend’s current native counts and cost estimator.

[8–9] Quantinuum Systems and Job Costing documentation (accessed 8 Oct 2026).

Make accuracy versus full cost the central result

$$\epsilon_E = |\widehat{E} - E_{\mathrm{ref}}|$$

MAIN FIGURE

Energy error + uncertainty
versus cumulative cost

SECOND FIGURE

Success probability
versus a fixed resource budget

H_2 is a validation case. Add geometries and larger active spaces before claiming scaling.

Four shortcuts that lead to wrong conclusions

“NO NORMALIZATION NEEDED”

The unnormalized vector can grow or vanish.

“A PLATEAU MEANS SUCCESS”

A wrong sector or restricted pool can also stall.

“LOCAL MEANS SHALLOW”

The full preparation circuit usually accumulates.

“BELOW FCI MEANS BETTER”

Check noise, conditioning, and the reference model.

Three checks before you move on

01 Explain

Why can negative energies give growing amplitudes while filtering still works?

02 Predict

For H = Z and $|+\rangle$, what happens when the pool contains only Z?

03 Audit

One method uses fewer shots but deeper circuits. Which resource comparison is missing?

The mental model to keep

01 The filter supplies the target direction.

02 The pool limits the available directions.

03 Measurements estimate a Gram system.

04 A classical solve chooses the unitary update.

05 **The full circuit and every measurement have a cost.**

Primary papers: read with a purpose

[1] Motta et al.

Local-unitary QITE and quantum Lanczos. Nature Physics 16, 205–210 (2020).

arXiv:1901.07653

[2–3] McArdle et al.; Yuan et al.

Variational imaginary-time evolution and its broader theory.

arXiv:1804.03023; Quantum 3, 191 (2019)

[4] Cortes & Gray

Quantum Krylov subspace algorithms for energy estimation.

arXiv:2109.06868

[5] Epperly, Lin & Nakatsukasa

Conditioning and thresholding in quantum subspace diagonalization.

arXiv:2110.07492

[1] Motta et al. | [2] McArdle et al.

Documentation and reproducible learning files

[6] OPENFERMION

Electronic-structure tutorial
Explicit H_2 Hamiltonian and conventions

[7] QISKIT ALGORITHMS

Variational time-evolution tutorial
A separate ansatz-based formulation

[8–9] QUANTINUUM

Selected-system workflow and costing
Verify device and software versions

YOUR COMPANION MATERIALS

PDF + self-contained LaTeX
Python tests, figures, and raw CSV data

[6] OpenFermion | [7] Qiskit Algorithms | [8–9] Quantinuum documentation